Week 5 Report

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Role: Frontend Developer & Scrum Master

Date: August 31, 2025

Project: OAS - Web-based Oral Assessments

Reporting Period: Monday, August 25, 2025 – Sunday, August 31, 2025

1. Tasks Completed This Week

Frontend Development & UI Implementation

- **Complete Student Dashboard System** (Evidence Link)
  - **Comprehensive Dashboard Implementation:**
    - Developed complete student portal layout with sidebar navigation and top navigation bar.
    - Implemented StudentStatsCards component displaying active assignments, upcoming assessments, and completed tasks.
    - Created StudentCalendar component with full calendar integration for assessment scheduling.
    - Built StudentQuickActions component providing rapid access to common student tasks.
    - Integrated RecentResults and UpcomingAssessments components with real-time data updates.
  - **Service Layer Architecture:**
    - Developed comprehensive studentService.js with mock data integration.
    - Implemented dashboard data aggregation and statistics calculation.
    - Added dark mode support through custom useDarkMode hook.
    - Created responsive layouts supporting mobile and desktop views.
- **Student Groups Management System** (Evidence Link, Evidence Link)
  - **Complete Groups UI Implementation:**
    - Implemented groups listing page at /student/groups with comprehensive filtering capabilities.
    - Created detailed group view page at /student/groups/[groupId] with member information.
    - Developed MyGroups dashboard component with enhanced data visualization.
    - Built group member cards with role indicators and contact information.
  - **Backend Integration Preparation:**

        - Created studentGroupsService.js with complete CRUD operations.
        - Documented Student-Groups-API-Contracts with detailed endpoint specifications.
        - Developed Student-View-My-Groups-User-Flow documentation.
        - Added implementation plan for future backend integration.
- **Student Results Viewing System** (Evidence Link)
    - **Comprehensive Results Interface:**
        - Implemented results listing page at /student/results with filtering and sorting.
        - Created detailed result view at /student/results/[assignmentId] with feedback display.
        - Developed ExaminerFeedback component with rich text rendering capabilities.
        - Built RubricBreakdown component with visual progress indicators.
        - Added ResultCard and ResultsTable components for flexible data display.
    - **Service Integration & Documentation:**
        - Created comprehensive studentResultsService.js with 550+ lines of mock data handling.
        - Documented Student-Results-API-Contracts with complete API specifications.
        - Developed Student-View-My-Results-User-Flow with detailed user journey mapping.
        - Implemented ResultsEmptyState component for enhanced user experience.
- **Examiner Schedule Enhancement System**
    - **Advanced Schedule Filtering & Views** (Evidence Link)
        - **Multi-View Schedule Implementation:** Implemented WeekSchedule, DaySchedule, and enhanced ExaminerCalendar with month/week/day view switching. Added .ics export functionality.
        - **Global Filter System:** Developed a scheduleFilterStore for global state, created a ScheduleFilters component, and replaced alert notifications with a toast system.
    - **Examiner Timeslot Self-Assignment Interface** (Evidence Link, Evidence Link)
        - **Complete Assignment Workflow:** Implemented a multi-step wizard including AssessmentSelectionStep, TimeslotSelectionCalendar, and SelectionSummary.
        - **Service Layer Enhancements:** Improved examinerTimeslotService.js, created useTimeslotAssignment and useExaminerAuth hooks for state management.

Documentation & Architecture

- **Comprehensive API & User Flow Documentation** (Evidence Link, Evidence Link, Evidence Link)
    - **Student Booking Documentation:** Created API contracts and user flow documents.
    - **Student Results Viewing Documentation:** Created API contracts and user flow documents.
    - **Student Groups Management Documentation:** Created API contracts and user flow documents.

Bug Fixes & UI Improvements

- **Authentication UI Enhancement** (Evidence Link)
  - Implemented password visibility toggle, enhanced form validation, improved responsive layouts, and added proper error handling for authentication failures.

Scrum Master Responsibilities

- **Sprint Coordination & Feature Planning:** Led Week 5 sprint planning, coordinated student portal feature development, and managed integration priorities.
- **Code Review & Quality Assurance:** Conducted comprehensive reviews for all student portal implementations and ensured component reusability standards.
- **Documentation Leadership:** Standardized API documentation format and created comprehensive user flow documentation for all new features.

2. Time Allocation (Approximation for Week 5)

- **Frontend Development:** 30 hours
  - Student Dashboard Implementation: 6 hours
  - Groups Management System: 6 hours
  - Results Viewing System: 6 hours
  - Examiner Schedule Enhancement: 12 hours
- **Documentation & Architecture:** 6 hours
  - API contract documentation: 3 hours
  - User flow documentation: 3 hours
- **Scrum Master Activities:** 4 hours
  - Sprint planning and coordination: 2 hours
  - Code reviews and technical guidance: 1 hour
  - Team communication and standups: 1 hour

Total: 40 hours

3. Key Issues Encountered & Resolutions/Mitigation

- **Complex Student Data Relationships**
  - **Issue:** Managing interconnected student data across the dashboard, groups, and results required careful state management.
  - **Resolution:** Implemented centralized service layers with proper data normalization.
  - **Mitigation:** Established clear data flow patterns and documented the service layer architecture.
- **Schedule Filtering Complexity**
  - **Issue:** Implementing global filters across multiple schedule views while maintaining performance was challenging.
  - **Resolution:** Created a global filter store and optimized component re-renders through memoization.
  - **Mitigation:** Documented filter architecture and planned for performance monitoring.
- **User Flow Integration Challenges**

    - **Issue:** Coordinating multiple user flows with consistent UX patterns required extensive planning.
    - **Resolution:** Created comprehensive user flow documentation before implementation.
    - **Mitigation:** Developed design system guidelines and a reusable UI component library.

4. Plan for Next Week (Week 6: September 1 - September 7, 2025)

- **Priority 1: Student Booking System Implementation**
    - Complete student timeslot booking interface with calendar integration.
    - Implement booking confirmation and cancellation workflows.
    - Add conflict detection for overlapping bookings.
    - Integrate with backend booking APIs once available.
- **Priority 2: Backend Integration for Student Features**
    - Complete backend integration for student dashboard data.
    - Implement real-time updates for groups and results.
    - Add WebSocket support for live notifications.
    - Validate all API contracts with the backend team.
- **Priority 3: Performance Optimization**
    - Optimize large dataset rendering in results tables.
    - Implement virtual scrolling for long lists.
    - Add caching strategies for frequently accessed data.
    - Conduct performance testing with realistic data volumes.
- **Scrum Master Focus Areas**
    - Coordinate integration testing sessions with the backend team.
    - Facilitate user acceptance testing preparation for the student portal.
    - Manage technical debt assessment and prioritization for sprint 6.